Word boundary markers: a cross-linguistic perspective

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Abstract

Word boundary markers are essential linguistic cues that facilitate word segmentation, recognition, and comprehension in continuous speech. This paper explores the role of word boundary markers across languages, focusing on their segmental, suprasegmental, and morphophonological manifestations. Drawing on examples from languages such as Albanian, Georgian, English, French, Finnish, Chinese, Thai, Burmese, and others, the study explores the variety of boundary markers and investigates how they relate to morphological typology. The findings suggest that the type and frequency of boundary markers are determined by the morphological structure of the language, with isolating, agglutinative, and fusional languages exhibiting distinct patterns. Isolating languages, which lack inflectional morphology, rely heavily on tones and 'initials' and 'finals' to signal word boundaries. Agglutinative languages, which use extensive affixation (each usually encoding only one meaning), employ vowel harmony and weak word accent. Fusional languages, which combine multiple morphemes (usually with several meanings) into single words with blurred boundaries, often use fixed or free accent systems, consonant clusters, and final obstruent devoicing. The paper concludes with implications for future research in the context of language contact and change.

Keywords: speech segmentation; morphological typology; phonology-morphology interface

1 Introduction

For the fact that speech is continuous, word boundary markers need to be discovered. Word segmentation is a prerequisite for word recognition and subsequent comprehension of connected speech. Consider phonologically identical sequences with and without a word boundary in them, e.g.,

(1)

Night rate
n**i**trate
Nye trait

These are all disyllables with the first syllable accented, and the ones underlined and in bold are all pronounced differently by native speakers because of juncture phenomena. The difference is in the pronunciation of /t/ (aspiration and release) and the length of the vowel in /naɪ/ (see Lehiste 1960). Jusczyk, Hohne, & Bauman (1999) find that ten-and-a- half- month-old infants are sensitive to these allophonic cues. Phonetic transcriptions and references to word boundaries are given in

(2):

(2) Interplay between the word boundary and pronunciation[1]

Word/Phrase	IPA Transcription	Notes
Night rate	/ˈnaɪtˌɹeɪt/	Two-word phrase, slight boundary
nitrate	/ˈnaɪtɹeɪt/	Single word, no internal boundary
Nye trait	/ˈnaɪ ˈtɹeɪt/	Two-word phrase with prosodic break

Besides this often-cited example in (2), consider a random phrase from the Albanian language: /sotnukpunoɪ/. The obvious question to ask is how native speakers can comprehend this sequence. The answer is by chunking it into three legitimate Albanian words: /sot/ ‘today’, /nuk/ ‘not’, and /punoɪ/ ‘I work’. The cues for word boundaries are phonotactic constraints of the Albanian language. Albanian does not allow word-initial *tn and *kp (Klippenstein 2010). Having this input implicitly, native speakers decode the sequence efficiently as /#sot#nuk#punoɪ#/ ‘I do not work today’. Consequently, phonotactics serves as a robust mechanism for speech segmentation in this context.

Cutler (2012) explains this with a phenomenon coined by her as “Native Listening”. “Native listening” refers to how people who are native speakers of a language process and understand spoken language using the specific knowledge and expectations they have developed over time from that language.

> In Korean, released final stops are not legitimate. In Dutch, final stops should be released, at least when words are spoken in isolation. Cho and McQueen (2006) gave Korean and Dutch listeners a phoneme detection task, where the stimuli were VC nonsense syllables ending with stops in which the release burst either was intact or had been removed. The Korean listeners detected the stop targets without the burst more rapidly, whereas the Dutch

[1] For more of such examples in English, see Lehiste, 1960: 19.

listeners detected those with the burst more rapidly. (Cutler, 2012: 58)

English words typically carry accent on the first syllable (Cutler & Carter, 1987). Among English vocabulary items - ignoring frequency - the most prevalent type is a disyllabic word with initial accent, such as *lettuce* or *baggage* (Carlson et al. 1985). When English listeners are exposed to ambiguous two-syllable sequences (e.g., *lettuce* vs. *let us*, *intense* vs. *in tents*), they tend to interpret the trochaic (initially stressed) forms as single words more frequently (Cutler & Butterfield, 1990). In contrast, French listeners presented with similarly ambiguous sequences (e.g., *bagage* vs. *bas gage*) show the opposite preference: they are more likely to perceive iambic (finally stressed) patterns as single words (Banel & Bacri, 1994). This behavior aligns with the general pattern of final accent in French standalone words (Fletcher, 1991).

Listeners of different languages use varying strategies for parsing speech. What constitutes a good boundary cue in one language may be irrelevant or misleading in another. This variation arises because languages differ in their phonological structure, rhythmic organization, and morphological patterns.

This study explores word boundary markers from a cross-linguistic perspective, emphasizing their role in maintaining the independence and internal structure of words. By analyzing languages with distinct form-meaning mappings, this study establishes common patterns and limitations in marker usage, arguing that these patterns are conditioned by a language's morphological type.

The paper proceeds as follows. In Section 2, I introduce a background on the terminology and offer a classification of word boundary markers into three categories: segmental, suprasegmental, and morphophonological. Section 3 is devoted to morphological typology, classifying languages into three main morphological types: isolating, agglutinative, and fusional, by focusing on specific form-meaning mapping in these languages, relevant to the boundary phenomenon discussed in the paper. I claim that each type exhibits distinct patterns in the use of word boundary markers. In section 4, the methodology of the study is outlined. Section 5 presents findings, the interplay between morphological type and word boundary markers is illustrated on Thai, Burmese, Turkish, Finnish, Albanian, and Georgian. Section 6 concludes.

2 Terminology and classification

In this paper, I adopt Haspelmath's (2023) definition of the word[2]: "A word is (i) a free morph, or (ii) a clitic, or (iii) a root or a compound possibly augmented by nonrequired affixes and augmented by required affixes if there are any" (p. 285).

In the study of phonology and speech processing, several concepts have been proposed to explain how listeners identify word boundaries in continuous speech. One foundational idea comes from Trubetzkoy (1939), who introduced the notion of word boundary markers, signals, or cues-features in speech that help to signal the end of one word and the beginning of another. These cues may include changes in pitch, lengthening of final syllables, or specific phonetic patterns that consistently occur at word boundaries.

Building on this idea, scholars such as Lehiste (1960) and Hockett (1955) focused on the concept of juncture, which refers to the transitions or pauses between sounds that can indicate boundaries between words. Juncture plays a key role in distinguishing between otherwise ambiguous sequences of sounds (e.g., *night rate* vs. *nitrate*).

[2] Tallman (2020) has an interesting proposition concerning the definition of term *word*: "…it is not simply a question of whether word or constituency tests tend to converge or diverge. We should not expect them to. We should ask whether the results of constituency diagnostics are such to support the idea that there is some latent variable underlying their patterning that implies discontinuity in the tightness of constituency structure from morphs to sentences. Addressing this question will require that studies focus on relating the word to a global assessment of constituency structure" (p.10).

In addition to these, researchers have identified word-edge phenomena, which encompass phonological or prosodic features that appear specifically at the beginnings or ends of words. These may include accent patterns, intonation changes, or segmental alternations that are conditioned by word boundaries (Selkirk, 1984a).

Trubetzkoy (1939) also uses the term *delimitative element*, whose primary function is to mark the limits of words in spoken language. The term 'word boundary marker' is self-explanatory, which is why it has been adopted in this paper. Three main types of word boundary markers are distinguished as follows: segmental, suprasegmental, and morphophonological[3]. They are discussed in turn below.

2.1 Segmental boundary markers

These are individual sounds (segments) that help signal the boundary between words. I discriminate between processes and occurrence/realization.

2.1.1 Processes

There are three different types of boundaries in this category: structurally conditioned types of neutralization, lengthening, and insertion.

Structurally conditioned types of neutralization (e.g., final obstruent devoicing in German, Georgian, Polish, Czech, Russian, etc.). For instance, in German *bad* [baːt] 'bath', *rad* [ʁaːt] 'wheel', in Czech *narod* [narot] 'people', *hrad* [hrat] 'castle', in Polish *brzeg* [bʐɛk] 'shore', in Georgian *kargad* [kargat] 'well'. Note that devoiced obstruents reclaim their voicing when suffixes

[3] Certain issues ignored for the sake of focus of the paper are: i) negative boundary signals, both phonemic and non-phonemic, individual, as well as group ones (e.g., phenomena like flapping in American English, *hd*, *ks*, *ps*, *lk* word-medially in Finnish) and ii) Polysynthetic languages (having long "sentence-words").

are attached to the words. For instance, German, *bad* [baːt] ‘bath’ vs *baden* [baːdən] ‘to bathe’. Other languages do not have this phenomenon, e.g., in Persian *medad* [meˈdɒːd] ‘pencil’, *asb* [æsb] ‘horse’, and in French *chaise* [ʃɛz] ‘chair’, *promenade* [pʁɔ.mə.nad] ‘walk’. Butskhrikidze (1994a) finds a correlation between word-final obstruent devoicing and fixed accent placement in fixed accent languages and claims that devoicing does not occur in languages with final fixed accent (e.g., in French and Persian). This is due to the physical prominence of the final syllables (high pitch, loudness, duration, etc.) in these languages, while in languages with no final fixed accent, devoicing/neutralization might occur (e.g., in Czech, Polish, and German) as the word-final syllable is less prominent. Here, segmental and suprasegmental word boundary markers are in complementary distribution. The issue is discussed further below in this paper.

Constituent-final lengthening is a phenomenon that appears to be widespread across languages and may even be universal (Vaissière, 1983). Some scholars have suggested that this effect could arise from general, nonlinguistic perceptual mechanisms, rather than purely linguistic ones. For instance, Woodrow (1909) demonstrated over a century ago that sequences of regular sounds varying in duration tend to be perceived iambically - that is, with the longer sound interpreted as final. Crosslinguistic experiments support the perceptual salience of final lengthening: listeners of English, Dutch, and French all benefited significantly from final lengthening cues in speech segmentation tasks, while none showed similar advantages for initial lengthening (Tyler & Cutler, 2009).

Non-phonemic sound insertion refers to the addition of a sound that does not change the meaning of a word. These insertions are often due to ease of articulation, rhythmic patterns, or phonotactic constraints and are generally considered phonetic or allophonic rather than phonemic. An example of non-phonemic sound insertion is “glottal stop” insertion before the vowels in the word-initial position. It is attested in German, the Southern dialects of Polish, Armenian, and Bohemian dialects of Czech. For instance, in German *Apfel* [ˈʔap.fəl] 'apple', *Essen* [ˈʔɛ.sən] 'to eat'. In some English dialects (e.g., Cockney), a glottal stop may appear between words like *"uh-oh"* [ʔʌʔoʊ].

2.1.2 Sound occurrences/realizations

Languages often have different phonotactic rules for word-initial and word-final positions compared to medial positions. Sound occurrences at the word edges function as word boundary markers. They are more easily perceived, remembered, or processed than sounds in the middle of words. I distinguish between four different types: consonant clusters, phonemic group signals (Trubetzkoy, 1971), single/individual sound occurrences, and sub-phonemic realizations of sounds.

The discussion of consonant clusters naturally calls for consideration of morphology. Dressler & Dziubalska-Kołaczyk (2006) propose an important distinction between morphonotactic (MPT) vs. phonotactic clusters (PT). They argue that morphonotactic clusters are those that arise phonologically, from vowel loss, and morphologically from concatenation, as opposed to phonotactic clusters that occur in monomorphemic context. Two important predictions made by the morphonotactics (a distinct subfield of morphonology), relevant to this paper, are the following: a) the stronger inflecting a language is the more morphonotactic clusters it should have (the morphologically richer Slavic languages have more MPT clusters than the morphologically poorer Germanic and Romance languages see also Dressler, et.al (2019) and b) prototypical morphonotactic clusters are always phonotactically marked, i.e. they are dispreferred with respect to comparable prototypical phonotactic preferences. These claims made in Dressler & Dziubalska-

Kołaczyk (2006) are substantiated in subsequent studies (e.g., in Kelić & Dressler (2019) and Dressler & Kononenko 2020) by linguistic data from Germanic, Slavic, Baltic, Romance, and later in this paper, by Albanian and Georgian (though going into detail is beyond the scope of this paper and requires separate in depth investigation).

There are constraints on the length and constituency of consonant clusters in the word-initial and word-final positions in any language[4]. For instance, in English, only a small set of CCC onsets is allowed. They typically follow the pattern: /s/ + voiceless stop (/p/, /t/, /k/) + approximants (/l/, /r/, /w/ and /j/). Some illustrative examples are given in (3).

(3)

/spl/ 'plash'
/spr/ 'spring'
/str/ 'street'
/skr/ 'scream'
/skw/ 'squash'

These are the only types of three-consonant onsets allowed in native English words.

Phonemic group signals (e.g., at a boundary between two units of meaning, e.g., consonant + h (in German *ein Haus* 'a house'), or nasalized vowel + m (in French). For more on these types of signals, see Trubetzkoy (1971: 290).

[4] For more on cross-linguistic generalizations concerning initial and final consonant clusters see Greenberg (1978).

Under single/individual sound occurrences, distribution restrictions at the word edges are implied: e.g., /h/ (word-initially)[5], and /ŋ/(word-finally) (in English). In Finnish, the glottal stop is the word-final marker (Suomi, K., Toivanen, J., & Ylitalo, R. (2008)). Many similar examples could be listed from any given language. For instance, as Booij (1983) says: "… languages have special restrictions for the word-initial and word-final positions that are not valid for every syllable onset and coda, respectively". He gives following examples of languages where restrictions hold word-finally, but not syllable-finally:

(4)

Word-final restrictions		Source
Japanese		Vennemann (1978)
Huichol		Bell (1976)
Timicua	no word-final consonants	Granberry (1956)
Campa Arawak		Dirks (1953)
Sierra Nahuat	no word-final [m]	Key & Key (1953)
Arekuna Carib	no word-final [m], [n]	Edwards (1978)
Spanish	no word-final [ns]	Pulgram (1970)

(Booij, 1983: 251)

And languages where restrictions hold word-initially, but not syllable-initially are given in (5):

(5)

[5] The generalizations presented in this paper pertain exclusively to Standard pronunciation. Nevertheless, /h/-dropping in word-initial position is attested in certain English dialects, e.g., those spoken in Yorkshire and Lancashire.

Word-initial restrictions		*Source*
Tamil	no word-initial lateral segments	Fowler (1954)
English	no word-initial [ž]	Hooper (1976:197)
Marihuana	no word-initial consonant clusters	Pike & Scott (1962)
Sierra Nahuat	no word-initial [h], [g]	Key & Key (1953)
Ongolo	no word-initial consonants (except certain prefixes)	Dixon (1970)
Dutch	no word-initial [pj], [kj]	Booij (1981)
Bamileke	no word-initial glottal stop	Hyman (1978)

(Booij, 1983: 252)

Sub-phonemic realizations of sounds, e.g., clear /l/ in English, occur word-initially (in contrast, dark /l/ is found word-finally). In Tamil, p^h, t^h, and k^h only occur word-initially; /g/ is attested only word-initially in Japanese[6].

It is important to note that all word boundary markers discussed in this section involve consonants. It is not an accident. “The traditional etymology of the Sanskrit name for consonant, vyañjana, as ‘revelative’ seems to carry the suggestion that the consonants rather than the vowels are responsible for the differentiation of meaning” (Allen 1953: 81, cited in Butskhrikidze 2002). Cross-linguistically, in phonemic inventories, consonants largely outnumber vowels, which indicates that consonants are better at discriminating meaning than vowels. “…the opportunity to turn one word of any language into another by replacing a consonant is, in general, much greater than the same chance with replacement of a vowel” (Cutler et al. 2000: 754).

There are many psycholinguistic experiments illustrating the consonant vs. vowel dichotomy in the studies on noise masking (Noteboom & Doodeman, 1984), slips of the ear (Bond & Garnes, 1980), and the word reconstruction task experiments (Ooijen, 1996). For instance,

[6] In native Japanese vocabulary, /g/’s allophonic variant in intervocalic position is [ŋ]. Vance (2008) discusses the phonemic status of /g/ in Japanese and its positional allophones, namely [g] and [ŋ], as well as their variation across generations. In addition, Hibiya (1995) addresses the decline of intervocalic [ŋ] among younger speakers in Tokyo, with a corresponding shift toward a more consistent realization as [g]. There are also regional differences in this respect, the discussion of which lies beyond the scope of this paper.

Ooijen (1996) claims that English listeners regard vowels as inherently mutable and as less reliable in constraining lexical access. In the experiment on lexical selection, listeners think of *kebra* as more like a *cobra* than like a *zebra*. For cross-linguistic comparisons on constraints of vowels and consonants on lexical selection, see Cutler et al. (2000).

Language acquisition studies also illustrate this dichotomy. Jakobson & Waugh (1979) propose that in children's language, the sense-determinative role of consonants generally antedates that of vowels (i.e., the oppositions within the consonant system appear before those in the vowel system).

When discussing consonant vs. vowel asymmetry, we could also refer to studies on writing systems and studies on tip-of-the-tongue experience, because how we retrieve words is related to how we encode them.

Focusing on the relevance of this asymmetry for phonotactic constraints, it is crucial to note the following observation from Butskhrikidze (2002: 61):

(6)

i. Vowel co-occurrence displays an associative tendency.
ii. Consonant co-occurrence displays a dissociative tendency[7].

Consonant combinations tend to be disharmonious. The principles of the Obligatory Contour Principle (the OCP (Leben 1973; McCarthy 1986)) and the Sonority Sequencing Principle (the SSP (Clements, 1990; Selkirk, 1984b) substantiate this generalization. The fact that vowel

[7] I will come back to the implications of the dichotomy in (6) later in this paper (see section 3).

harmony is cross-linguistically a more widespread process than consonant harmony is also attributed to this observation.

As an alternative to these phonological principles, Dziubalska-Kołaczyk 2009, 2014, 2019 suggest the Net Auditory Distance principle (NAD)[8] providing a more comprehensive and detailed account of the phonological structure of clusters than the sonority principle. The NAD is grounded in Natural Phonology (Donegan and e 1979; Donegan and Nathan 2015), and Beats-and-Binding phonotactics (Dziubalska-Kołaczyk 2002, 2003) which focuses on perceptual and articulatory naturalness[9]. NAD is calculated using manner of articulation (MOA), place of articulation (POA), and the sonorant/obstruent (S/O) distinction. Each of these features is assigned numerical values (e.g., different MOAs and POAs get distance scores), and the sum gives a "distance" between two consonants. The principle states *well-formedness conditions* in clusters depending on their position in the word, for example, in initial C1C2V clusters, NAD (C1, C2) should be greater than or equal to NAD (C2, V). For a concrete application of this principle on Polish and Italian consonant clusters see Bertinetto, et al. (2007). Though I see a promising application on the principle in the fusional languages discussed in this paper, it is outside the scope of the present paper and should be explored in a separate study.

2.2 Suprasegmentals/word level prosody

[8] The NAD principle has been extended and revised in Orzechowska & Dziubalska-Kołaczyk (2015) and Orzechowska & Dziubalska-Kołaczyk (2023).

[9] Steriade (1999) also claims that consonantal phonotactics are best understood as syllable-independent, string-based conditions reflecting positional differences in the perceptibility of contrasts.

These involve features above the level of individual sounds, such as accent, intonation, rhythm, or pitch. I focus on three different categories: free accent (Spanish, English, Dutch, Albanian, Russian); fixed/predictable accent (word-initial (Estonian, German[10] (in the native vocabulary), Icelandic, and Czech), word-final (Persian and French), and penultimate (Polish)); and Tones (Chinese, Burmese, Vietnamese). I examine different functions of suprasegmentals: a) distinctive, b) delimitative, or c) both, in these languages.

In free accent languages, accent seems more distinctive than delimitative[11]: illustrative examples of this are word pairs such as: English *REcord* (noun) vs. *reCORD* (verb), *IMport* (noun) vs. *imPORT* (verb), and accent falling on different syllables in words, e.g., *languages* [ˈlæŋ.gwɪdʒ] (word-initially), *computer* [kəmˈpjuːtə] (in the middle of the word) and *refuse* [rɪˈfjuːz] (word-finally). A characteristic of this type of language is accent shift in derived words. For instance, in English, *photograph* [ˈfəʊ.tə.grɑːf], *photographer* [fəˈtɒ.grə.fə] and *photographic* [ˌfəʊ.təˈgræ.fɪk]. In this example, a shift in accent placement from the first to the second and then to the third syllable of the derived word is attested.

A fixed-accent language is one where the position of accent in a word is predictable; it always falls on a certain syllable (e.g., the first, last, or penultimate). Hyman (1997) claims that, "Limiting our attention to demarcative stress, i.e., to stress which signals a word boundary, it should be clear that the closer stress falls to that boundary, the better it will fulfill its linguistic

[10] In German, primary word accent usually falls on the first syllable of the word stem (or root) for native vocabulary. However, there are varied opinions on word accent in German, e.g. Hulst (2010) suggest that German has a quantity sensitive system with the following accent assignment rules: "Accent is final if the final syllable is superheavy (VVC, VCC) b. If the final syllable is open accent is penultimate c. If the final syllable is closed (but not superheavy) and the penultimate syllable is open, accent is on the antepenultimate syllable (s´ CV CVC) (Hulst, 2010: 447).

[11] Accent pattern being delimitative in English could be illustrated by the example: *blackbird* (compound noun), accent on the first syllable vs. *black bird"* (adjective + noun) accent on both words or the second word more. It helps distinguish between a single word and a word boundary.

function. And not surprisingly, initial and final stress are commonly attested in the world's languages” (p. 41).

Fixed accent does not change to mark differences in meaning between words. Predictable accent helps both speakers and listeners process words more efficiently.

Listeners can anticipate where emphasis will fall and recognize word boundaries more easily. Consider an example from German: *Mein Freund kommt heute* [maɪ̯nˈfʁɔʏ̯ntˈkɔmtˈhɔɪ̯tə] ‘My friend is coming today’ in which each word is identifiable by initial word- accent.

The correlations between the fixed accent placement and other word boundary markers, e.g., final obstruent devoicing and consonant clusters, are explored in Butskhrikidze (1998).

Unlike free accent, fixed accent is strictly delimitative.

Over 60% of the world’s languages use tone to some extent (Yip, 2002). I adopt the definition of tone suggested by Hyman (2018): “A language with tone is one in which pitch is a contrastive feature of at least some morphemes” (p.698). They are common in East and Southeast Asia (e.g., Chinese, Thai, Vietnamese), Africa (e.g., Yoruba, Igbo, Zulu), and the Americas (e.g., Navajo, Mixtec). In many tonal languages, each syllable or mora (sub-syllabic timing unit) carries one tone. Some languages allow toneless syllables or floating tones that attach to nearby syllables. Tones in Southeast Asian languages are primarily lexical, using distinct pitch contours or voice quality to differentiate meaning, while tones in many Bantu languages are mostly grammatical and primarily use a simpler two-level (high/low) system, though some African languages also have complex tonal systems and down-step. Southeast Asian tone systems have a greater variety in pitch, with some featuring up to eight tones, and often correlate with features like a creaky or breathy voice, whereas Bantu languages more frequently rely on tone for grammatical functions rather than purely lexical distinction. Given the focus on word boundary markers, the scope of this investigation is restricted to languages featuring lexical tones. Furthermore, any subsequent use of the term *tone* refers exclusively to lexical tone. Tone is phonemic: a difference in pitch alone can create different words**.** An example from Mandarin Chinese is given in (7):

(7)

mā (妈 'mother')

má (麻 'hemp')

mǎ (马 'horse')

mà (骂 'to scold')

Tone may assist in identifying word boundaries in tonal languages, but this function is usually secondary to its primary role of distinguishing word meaning or grammatical information. Lexical tone patterns can signal word edges. In some tonal languages, specific tone sequences are more likely within a word than across word boundaries**.** Listeners can learn these tone co-occurrence patterns and use them to infer where words begin and end**.** For instance, Yoruba has High (H), Mid (M), and Low (L) tones. Native speakers can often predict likely tone sequences in compounds vs. phrases, e.g., within a word: HL is common; across boundaries, unexpected HL patterns might cue segmentation.

Tone sandhi can also reveal word boundaries. In languages with tone sandhi (tone changes triggered by phonological context), changes often occur at word boundaries**.** This makes tone alternations a signal for where words start or end. The following is an example from Mandarin Chinese:

(8)

Third Tone Sandhi: When two low (third tone) syllables occur together, the first one becomes rising (second tone).

nǐ + hǎo → ní hǎo

Speakers use this sandhi rule to anticipate syllable or word boundaries in rapid speech[12] (Lai and Li 2022, Chen, and Yuan, 2007).

Additionally, in some tonal languages, intonation patterns (phrase-level pitch movement) are layered on top of lexical tone. The way pitch resets or drops at phrase boundaries can indirectly help listeners detect word or phrase breaks. Thai uses both lexical tone and intonation contours. At the end of a phrase, there might be a reset in pitch (Slayden, 2009), hinting at the start of a new unit (possibly a new word or phrase). The example is taken from (Slayden, 2009: 10-11).

(9)

[12] It is interesting to compare this suprasegmental phenomenon to the similar segmental phenomenon, called liaison. Liaison in French can help listeners identify word boundaries during speech recognition, but it does so in a somewhat paradoxical way. French liaison simultaneously hides and signals word boundaries. Liaison consonants are acoustically different from ordinary consonants. Listeners unconsciously use these duration cues to infer word boundaries (Spinelli, Cutler, and McQueen 2002).

$$/\text{tone-33}/_{\text{syllable}} \rightarrow [\text{tone-32}] \; / \; \_\_]_{\text{phrase}}$$

At the end of a phrase, the mid tone is dropped very slightly.

/set^{11} lɛːw^{45} rɯː^{24} jaŋ^{33}/	[... jaŋ^{32}]	เสร็จแล้วหรือยัง	“Are you finished?”

$$[\text{tone-45}]_{\text{syllable}} \rightarrow \begin{bmatrix}\text{tone-454}\\ +\text{low}\end{bmatrix} \; / \; \_\_]_{\text{phrase}}$$

At the end of a phrase, the high tone has glottal stricture and drops quickly.

$$[\text{tone-52}]_{\text{syllable}} \rightarrow [+\text{low}] \; / \; \_\_]_{\text{phrase}}$$

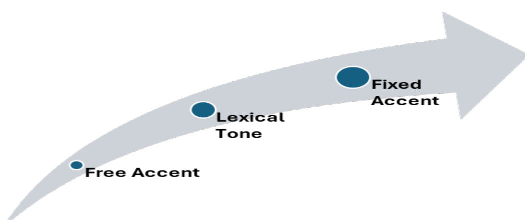
At the end of a phrase, the falling tone has glottal stricture.

/tɕʰan^{24} tɔŋ^{52} kaːn^{33} ma^{45} la^{45} kɔ̄ː jaj^{11} nɯŋ^{11} luːk^{52}/	[...luːkˀ]	ฉันต้องการมะละกอ ใหญ่หนึ่งลูก	“I want one large papaya”

[13]

When comparing free accent, fixed accent, and tone in terms of how effectively they mark word boundaries, I propose the following continuum depicted in (10):

(10) The strength of suprasegmentals as word boundary markers



To my knowledge, no experimental studies have compared the strength of fixed accent, free accent, and tone as word boundary markers by measuring their effectiveness e.g., word spotting or word recognition in continuous speech. The suggestions presented in (10) are purely analytical. Fixed accent systems (e.g., Czech or Slovak) demonstrate the highest effectiveness because the accent, typically associated with high pitch, duration, or intensity, is always anchored to a specific syllable within every word. Thus, it functions as an automatic, absolute boundary marker. Conversely, the

[13] For tones, following notation are applied in this example: ^{11}-low, ^{33}-mid, ^{45}-high, ^{24}-rising and ^{52}-falling.

effectiveness of tones (e.g., Mandarin or Yoruba) varies. Tone is lexically contrastive, meaning it distinguishes word meanings rather than grouping syllables into words. Thus, it has paradigmatic function, rather than syntagmatic (e.g., like fixed accent). However, in tone sandhi systems, tonal changes can indirectly serve as boundary cues. Finally, free or variable accent systems (e.g., Russian) exhibit the lowest effectiveness. Because the accent can fall anywhere within a word, it offers minimal boundary information unless paired with vowel reduction or rhythmic patterns (such as iambic versus trochaic foot structures) to signal the beginning of a new prosodic unit. In this regard, an interesting experimental study conducted by Zubov and Riekhakaynen (2024) on Russian focused on the assessment of words with variable stress versus those with only one standard stress pattern. They found that processing the correct pronunciation of words with variable stress takes significantly more time than evaluating words that possess only a single correct pronunciation variant.

Accordingly, fixed accent serves as the strongest marker of word boundaries, free accent as the weakest, and lexical tone occupies an intermediate position.

2.3 Morphophonological boundary markers

Morphophonological word boundary markers are sound or form-based cues that help identify where one word finishes and the next begins, based on the interplay between morphology (word structure) and phonology (sound patterns). This study only focuses on vowel harmony as a word boundary marker[14].

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[14] Some languages allow harmony to spread across clitics or even multiple words in fluent speech. Thus, it can't always be trusted to mark word edges.

Vowel harmony is a phonological process where vowels within a word must agree in certain features (e.g., frontness, backness, roundness, or height). In many vowel harmony languages, the rule only applies within words, not between them. Consequently, when harmony stops, that signals a possible word boundary. Turkish has backness vowel harmony:

(11)

ev ‘house’ *ev-ler* ‘houses’

okul ‘school’ *okul-lar* ‘schools’

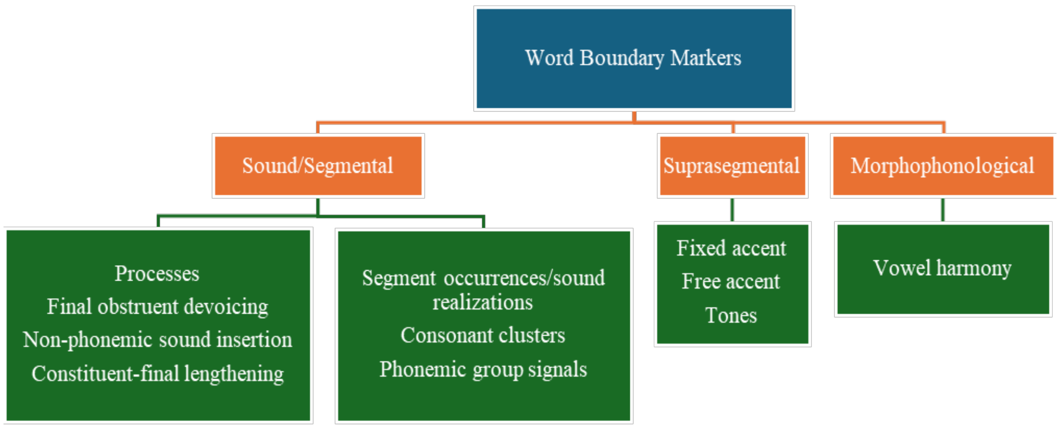
But in multi-word sequences, harmony does not apply across words.

Thus, if there is a disharmonic vowel after a harmonic sequence, it may signal the start of a new word. E.g., in a phrase *evler açık* [evˈleɾ aˈtʃɯk] ‘houses are open’, there is no harmony across the words**.** The disharmony between *-ler* and *açık* helps signal a word boundary. Consider another example from the Finnish phrase in which disharmony signals word boundaries, e.g., *Kylä on hyvä* [ˈkylæ on ˈhyvæ] ‘the village is good’. Finnish also has backness harmony, which does not cross word boundaries. In the given phrase, the first word *Kylä* has front vowels, the second *on* - back vowel, while the third *hyvä* - front vowels. For experimental work on how vowel harmony is exploited by listeners for segmentation of continuous speech, see Suomi, MacQueen & Cutler (1997).

2.4 Summary

All word boundary markers discussed above are summarized in (12).

(12) Classification of word boundary markers



The figure in (12) depicts a wide array of word boundary markers that exist theoretically. In this paper, I argue that only a limited set of boundary markers are attested in a specific language.

Across a series of studies, Butskhrikidze (1994a, 1996a, 1996b, 1998) investigated the interplay of word-boundary markers in 55 languages characterized by fixed accent, noting that these markers can either mutually exclude or accompany one another. Butskhrikidze (1998) established three primary correlations regarding word boundaries. First, fixed accent placement and word final obstruent devoicing rarely co-occur; languages lacking final devoicing typically feature word final fixed accent (e.g., French, Persian), whereas languages with penultimate or antepenultimate accent often have final devoicing (e.g., Polish, German). Second, the position of fixed accent tends to match the position of obstruent clusters, meaning word-initial accent aligns with initial obstruent clusters (e.g., Czech, Georgian), while word-final stress aligns with final clusters (e.g., French, Grabar). Third, morphological structure of citation forms influence accent assignment, where languages with citation forms coinciding with stems correspond to word final fixed accent, while languages with citation forms represented as stem + affix, correlate with non-final fixed accent. The last correlation indicates that morphology plays a crucial role in predicting accent placement, i.e., what boundary markers to expect in a language.

Consequently, when asking a question, what factors limit the use of specific word boundary markers in a language, and to what extent can their distribution be predicted? A short answer is morphology**:** types of form-meaning mapping (one-to-one or one-to-many or many-to-one) a language possesses; the way radical and relational (Sapir, 1921) ideas are expressed in a language. A more elaborate discussion on morphological typology and its relevance to word demarcation is given in the following section.

3 Morphological typology

Absolute linguistic universals are features that are found in every known human language without exception. They are very rare and often debated. However, these two in 13(i) and 13(ii) stand out, are interrelated, and directly tie into the issue of boundary markers discussed in the paper.

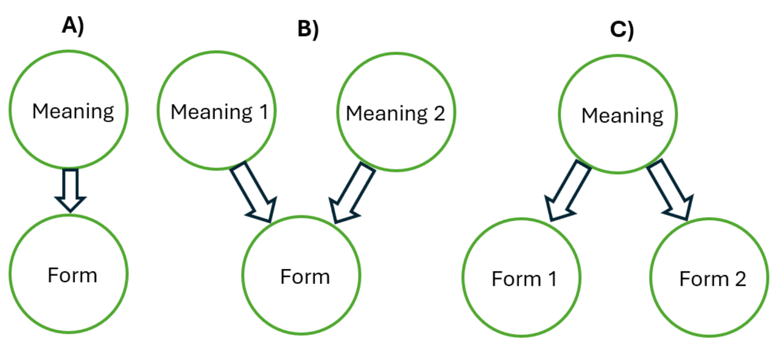
(13)

i. All spoken languages have vocabulary and grammar
ii. All spoken languages have consonants and vowels in their phonemic inventory

When you tease parts from the whole, you create tension. These two universals exhibit that tension and are interconnected. The most vivid illustration of the interdependence between 13(i) and 13(ii) is in Semitic languages: consonants - vocabulary, vowels - grammar[15]. Additionally, one more dichotomy relevant to the discussion is that words are generally discrete units, while phrases are continuous. However, the issue is more complex than it initially appears, as it largely hinges on the nature of form-meaning associations in affixes within a given language.

I suggest, there are three scenarios of form-meaning mappings for affixes in a language depicted in (14).

(14) Types of form-meaning mappings (Illustration of one-to-one, many-to-one, and one-to-many mappings in affixes, showing how isolating/agglutinative vs. fusional languages diverge).



The type of mapping depicted in (14A) is attested in isolating languages**[16]**; the example in (7 from Mandarin Chinese illustrates one form - one meaning mapping. Type (14B) mapping can be

[15] Butskhrikidze (2002) claims, "One finds few languages with obstruent clusters as constituents of a grammatical morpheme. If a language has consonant clusters, they are more likely to appear within a stem, rather than within a grammatical morpheme. This correlation between the stem and obstruent clusters is attested in fixed accent languages (Butskhrikidze 1998)" (p.9).

[16] One could consider, instead of using the traditional isolating-agglutinative-fusional typology, the language types be framed in terms of the indices of synthesis and fusion. Adopting this approach, however, would require assigning specific numerical values to all six languages discussed in this paper. While calculating the index of synthesis is relatively straightforward, determining the index of fusion is considerably more complex due to phenomena such as portmanteau morphemes, suppletion, and autosegmental morphology, among others. Consequently, this approach may introduce more ambiguity than clarity, as there is substantial disagreement concerning the classification of certain

illustrated on Modern Greek, e.g,. *γráfete* ‘was being written’ where the -*ete* suffix has five different meanings: 3rd person, singular, passive voice, durative and past tense. Such a type of morpheme is often referred to as a portmanteau morpheme. Type (14C) mapping can be illustrated on the expression of plurality in English. There are many ways of expressing plural, e.g., by adding -*s* or -*es* to a noun (e.g., *book-s* [bʊks], *box-es* [bɒksɪz]), changing a vowel (e.g., *man* [mæn] → *men* [men]), etc.

Type (14C) mapping in affixes is also attested in agglutinative languages since vowel harmony adjusts affixal vowel to a stem vowel. Nevertheless agglutinative languages have predominantly (14A) type mapping. Consider a phrase from an agglutinative language, Buriat (Mongolian) in (15):

(15)

jaba-na-gyi-b
go PRES NEG 1SG
‘I do not go’ (Poppe, 1960: 57)

Because of a one-to-one mapping between form and meaning, it is easy to find morpheme boundaries in this sequence. Additionally consider, for instance, the ablative plural of the Turkish word *ev* ‘house’ *ev- ler-den* ‘house PL ABL ‘from houses’.

Languages with a high frequency of Types (14B) and (14C) are characterized by fusion. Words/morphemes in these types of languages is difficult to separate in a continuous speech. For instance, sandhi in East Norwegian in the form [sva:ʈ̆] 'answered', where the verb root [sva:r] is

morphemes. For instance, the Georgian thematic suffix -*eb* in the derived word *a-k’et-eb-in-eb* ‘you get X to make Y’. For a more detailed discussion of these indices and their implications for language description, see Payne (2017).

combined with the suffix [-t]. In this past participle form [r] + [t] merge into the retroflex [ʈ], fusing the two morphemes together. Arabic plural form *rijālun* of *rajulun* 'man' is another example of fusion (see more examples from Albanian and Georgian in section 5.3).

Based on the form-meaning mapping tendencies in different languages, it is apparent that, in languages where form-meaning mapping is one-to-one, morphemes are independent, while in languages having predominantly (14B) and (14C) types of mappings, morphemes are dependent.

I introduce four logical possibilities of classifying any language[17] according to the formal independence of lexical morphemes in citation forms and the formal independence of grammatical morphemes in (16).

(16) Logical possibilities of classifying any language according to the formal independence of lexical and grammatical morphemes

i. #lexical morpheme# = #citation form#
   grammatical morpheme - free

ii. #lexical morpheme# = #citation form#
   grammatical morpheme - bound

iii. #lexical morpheme# ≠ #citation form#
   grammatical morpheme - bound

iv. #lexical morpheme# ≠ #citation form#
   grammatical morpheme - free

The formal independence of lexical and grammatical morphemes in these four types of languages is summarised and illustrated by languages/language families in (17):

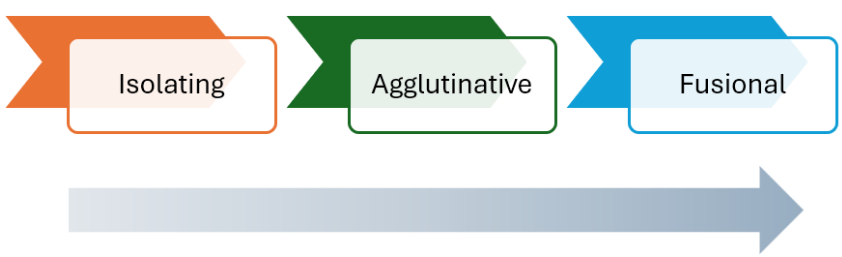
(17) The classification of languages according to morpheme types

[17] When classifying languages, we acknowledge that no language is entirely 'pure' of a particular type; instead, languages exist along a continuous spectrum.

Type of a morpheme Type of a language	Lexical morphemes	Grammatical morphemes	Languages/language families
i.	Independent	Independent	Chinese, Burmese, Vietnamese
ii.	Independent	Dependent	Altaic languages, North Caucasian languages, Turkish, Hungarian, English, French, Persian, Grabar
iii.	Dependent	Dependent	Most Indo-European languages (Polish, Russian, Albanian, Chez, German), South Caucasian languages (Georgian, Svan) Arabic Bantu (e.g., Zulu)
iv.	Dependent	Independent	Not known to me any language exemplifying this type[18]

As fusion increases, linguistic units lose their independence and the need for boundary markers increases. This is depicted in (18):

(18) The increase of blurring of formal boundaries from isolating to fusional languages



It is hypothesized that along the morphological continuum spanning from isolating to fusional languages (with agglutinative structures situated intermediately), a progressive blurring of word boundaries occurs, necessitating a corresponding increase in word boundary markers.

[18] This gap in the table is indicative that lexical units are inherently discrete and independent, while grammatical-dependent.

Consequently, isolating languages are anticipated to exhibit the lowest variety of word boundary markers, whereas fusional languages are expected to display the most diverse and frequent markers. Agglutinative languages show their distinct characteristics. These correlations are tested and explored in section 5. Remarkably, Dressler et al. (2019) suggests the following hypothesis: "The richer the morphology of a language is, the more MPT clusters will arise. This generalisation is limited by a typological variable (in the sense of Skalička 1979): inflecting-fusional languages allow more morphotactic opacity than agglutinating languages" (p.87).

In an empirical cross-linguistic study, Easterday et al. (2021) investigate the correlation between syllable complexity and morphological synthesis, offering a diachronic model that directly informs the typology of word boundary markers. Easterday et al. (2021) identify the diachronic pathway of grammaticalization as the driving mechanism. As free lexical elements undergo semantic reduction and morphosyntactic fusion, they experience corresponding phonetic erosion, often stripping down to single consonants. When these remaining consonantal affixes or clitics attach to a host word, they create complex consonant clusters precisely at the word boundaries. Based on the corpus of 21 diverse languages, the authors demonstrate that word-initial onset complexity correlates with synthesis at the beginning of phonological words (driven by prefixes), whereas word-final coda complexity correlates with synthesis at the end of phonological words (driven by suffixes). This positive relationship is explicitly concentrated within noun and verb structures. This study provides a foundational framework for understanding how grammatical markers become physically embedded within the phonotactic architecture of word boundaries, demonstrating that the structural complexity of an edge marker is a predictable consequence of historical language change.

4 Methodology

In Section 2, I introduced three types of boundary markers: segmental, suprasegmental and morphophonological. I proposed that some boundary markers can cooccur in a language, while others are in complementary distribution. Every language allows only limited number of boundary markers, and I suggested that specific word boundaries can be predicted based on the morphological type of a language. In section 3, I also explored different possibilities of form-meaning mappings in different morphological types: isolating, agglutinative and fusional and proposed that we can expect wider variety of word boundaries in fusional languages to compensate dependent nature of words/morphemes in this type of language. In Isolating languages, we expect the least variety as words/morphemes are independent, while the agglutinative languages will show an intermediate state (having independent stem, but dependent affix).

Having established the typological framework, we now test the predictions on six unrelated languages. I chose two representative languages from each morphological type. I tried to include language that are not genetically and geographically related. Six languages are discussed in the following section: Thai and Burmese (Isolating); Finnish and Turkish (Agglutinative), and Albanian and Georgian (Fusional). The data from Georgian and Albanian languages were obtained through fieldwork and phonetic and phonological studies in Georgia, Kosovo and Albania. The data from Thai and Burmese were drawn from grammar books and my previous investigations/published articles examining morphological, phonological and prosodic aspects. Data on Finnish and Turkish were collected from my studies into the phenomenon of vowel harmony. Additionally, first-hand knowledge of some of these languages, complemented by relevant literature, informed the analysis of specific linguistic phenomena.

Because this study analyzes a sample of six languages, the quantitative approach employed herein is strictly exploratory rather than confirmatory. This sample size is not intended to support sweeping typological generalizations or to definitively validate universal hypotheses. Instead, the current design serves as a targeted framework to identify preliminary cross-linguistic trends and to demonstrate the viability of this analytical model. Consequently, the statistical patterns reported in the following sections are interpreted as indicative tendencies rather than absolute quantitative claims, paving the way for larger-scale future replication.

5 Findings

I offer the following simplification to illustrate the difference between the three types of languages: isolating, agglutinative, and fusional, depicted in (19).

(19) Verifiably attested morphological types with illustrative examples

Isolating	#lexical morpheme# = #word# #grammatical morpheme# = #word# (e.g., Thai, Vietnamese, Burmese, Chinese)
Agglutinative[19]	#lexical morpheme# = #word# #grammatical morpheme# ≠ #word# (e.g., Altaic languages, some North Caucasian languages)
Fusional	#lexical morpheme# ≠ #word# #grammatical morpheme# ≠ #word# (Most of the Indo-European languages, Kartvelian/South Caucasian languages[20], Semitic languages)

[19] An interesting exception to this generalization is the Bantu languages, where #lexical morpheme# ≠ #word#. Typical word structure in this group is: prefix + stem, e.g., in Zulu and Shona. This obviously has implications for boundary markers in these languages. We leave the issue for future research.

[20] The Kartvelian languages (Georgian, Svan, and Megrelian-Laz) are often referred to as agglutinative languages. Melikishvili (2010) convincingly argues that Georgian and Svan are fusional. She illustrates it by examining verbal and nominal morphology with high irregularity, polysemy, ablaut and progressive, and regressive phonetic processes in both languages.

For the sake of brevity, we only consider two representative languages from each group: Thai and Burmese (Isolating); Finnish and Turkish (Agglutinative), and Albanian and Georgian (Fusional).

5.1 Isolating Languages

Key characteristics of Isolating Languages are the following: a) minimal or no inflection (words do not change form based on tense, case, number, gender, etc.); b) fixed word order (since there is no inflection to mark relationships, word order is crucial, and c) grammatical meaning is expressed through separate words (instead of changing a word to indicate tense or possession, separate words are used). Since grammatical relationships in isolating languages are expressed through independent words, an expanded vocabulary is anticipated, alongside the paradigmatic function of tone as a distinct differentiator of meaning. Lexical tones ensure the semantic differentiation of segmentally identical monosyllables, while 'initials' and 'finals' unequivocally demarcate word boundaries. Within this linguistic typology, final devoicing is unexpected because word-initial and word-final sounds are in complementary distribution, leaving no condition for final devoicing to occur. Furthermore, the strict assignment of initials and finals, combined with the absence of prefixation and suffixation, precludes the possibility of consonant clustering at word edges. Consequently, lexical tones, 'initials', and 'finals' are expected as primary word boundary markers in isolating languages. Two representative languages of this group are Thai and Burmese, discussed in turn.

5.1.1 Thai

Thai is part of the Tai-Kadai language family. It is closely related to Lao, Shan (spoken in Myanmar), and other Tai languages spoken across Southeast Asia, and is the official language of

Thailand**.** Thai is a tonal language with five tones: mid, low, falling, high, and rising. Consider five contrastive tones in Thai:

(20)

The mid tone	มยั	/māj/ ‘mule’
the rising tone	ไหม	/mǎj/ ‘silk’
the low tone	ใหม	/màj/ ‘new’
the falling tone	ไหม	mâj/ ‘burn’
the high tone	มยั้	/máj/ [particle]

(Sleyden, 2009: 5-6)

As it is demonstrated in (20), meaning bearing units are syllable sized.

> Most experts agree that Thai has twenty consonant phonemes (Abramson, 1962; Plam, 1990; Diller, 2008, p. 32). All consonant phonemes contrast in word/syllable/ –initial position … Only ten out of 20 can appear word/syllable finally: unvoiced oral stops /p t k/, nasals /m n ŋ/ approximants /w j/ and fricatives /s f/ (attested only in loanwords) (Iwasaki – Ingkaphirom, 2005, p. 4). Neither voicing nor aspiration contrast is attested in the syllable-final position. (Butskhrikidze 2021:829).

Diller (2008) claims that “Concerning the vowel phonemes, eighteen monophthongs (including nine short and nine long) are attested. Compound vowel phonemes (the combination of long monophthongs with /a/) in closed and open syllables are also attested. Interestingly short monophthongs do not allow such a combination” p. 33.

According to Iwasaki (2005), the structure of a maximal syllable is the following:

(21) The structure of a maximal syllable in Thai

(C_1) (C_2) V^T (/a/) (C_3)
C_1 = any consonant
C_2 = /w/, /l/, /r/
V = any monophthong
T = Tone
C_3 = /p/, /t/, /k/, /m/, /n/ /ŋ/, /j/, /w/

Iwasaki (2005)

As seen from the scheme in (21), initial consonants are optional, but in case both are present, their co-occurrence is constrained. Abramson (1962: 11-12) suggests the following constraint on the initial consonant clusters: C1 {p p^h t t^h k k^h} and C2 {l r w} (t^hr being rare). The approximant /w/ has limited combinatory power; it combines only with velar k and k^h. On the cluster simplifications in some of the Thai sub-dialects, see Slayden (2009).

There is an interesting experimental study by Winskel (2011) that makes the following claim:

> These results indicate that tone is playing an important role in early word recognition in Thai. The tone markers could also be functioning as salient word or syllable segmentation cues (e.g., หน้าต่าง/na:2ta:N/ “window”), as Thai does not have interword spaces to serve this function (Winskel, Radach, & Luksaneeyanawin, 2009). These findings concur with results from Chinese that tonal information is an essential component for word processing (Xu et al. 1999). When consonant and vowel information is activated in conjunction with tonal information, then this dramatically narrows the number of items potentially activated in tonal languages. Lexical tone in Thai is important as it can change the meaning of the word, as typically there are many syllables that are phonologically the same apart from the assigned tone. (p.754)

Consequently, Thai is characterized by lexical tones suprasegmentally, but possesses rigid, simple phonotactic rules at the word level.

5.1.2 Burmese

Burmese, also known as Myanmar, belongs to the Sino-Tibetan language group and is spoken in Burma. Burmese is an isolating language with some agglutination present in suffixes.

Watkins (2009) claims that “Morphemes in Burmese are predominantly monosyllabic. Except from Indo-European loans, typical from Pali and English, compounding is the major source for polymorphemic words” p.173.

Green (1995) claims that Burmese has four tones: high, low, creaky and checked. According to Green (1995), the syllable structure of Burmese is C(G)V((V)C), the onset consists of a consonant optionally followed by a glide, and the rime consists of a monophthong alone; a monophthong with a consonant or a diphthong with a consonant. Some illustrative examples come from Green (1995):

(22) Basic syllables of Burmese

a.	CV	mè:	‘girl’
b.	CVC	mɛ	‘crave’
c.	CGV	mjè:	‘earth’
d.	CGVC	mjɛ	‘eye’
e.	CVVC	màuN	(term of address for young men)
f.	CGVVC	mjáuN	‘ditch’

Green 1995: 68

> Regarding the phonotactic restrictions within morphemes, they are quite limited. Within a morpheme, neither combinations of two consonants nor combinations of two vowels are permitted. (The phonemes transcribed by “consonant + y and consonant + w” are palatalized or rounded consonants. hl, hm, etc., represent voiceless l' and m', and the diphthongs are to be considered monophonemic). The only admissible monomorphemic combination is the combination “consonantal phoneme + vowel phoneme” Trubetzkoy (1971: 246-247) (cited in Butskhrikidze 2021: 828).

Butskhrikidze (2022) argues that the uniformity of Burmese word types is related to the limited combinatory possibilities in this language.

5.1.3 Summary

Looking at word boundary markers in these two isolating languages, Thai and Burmese, the following can be observed. What serves as word demarcation in these languages are tones and ‘initials’ and ‘finals’. Thus, the following can be concluded:

(23) Word boundary markers in isolating languages

- ✓ Tones
- ✓ Initials and Finals (separate sound sets in different syllabic positions)
- ❑ No consonant co-occurrence restrictions in words
- ❑ Limited obstruent clustering, no devoicing, no stress, no vowel harmony

- ❖ Interesting in this respect is investigating tonogenesis, for instance, if losing voicing contrast is translated into the occurrence of different tones, more specifically, if voiced is turned into a high tone and voiceless into a low tone.

Next, agglutinative languages, Turkish and Finnish are considered in turn.

5.2 Agglutinative languages

Key characteristics of agglutinative languages are the following: a) clear morpheme boundaries, b) words made up of several morphemes (roots + affixes), c) each morpheme having a single, consistent meaning or grammatical function (morphemes are "glued" together (hence *agglutinative* from Latin *agglutinare*, 'to glue to'), d) morphemes are usually applied in a consistent order and form**.** Irregular forms are rare. Agglutinative languages exemplify a perfect balance between the lexicon and grammar. Both lexical and grammatical morphemes are easily identified due to a predominant one-to-one mapping between form and meaning. This structural balance allows vowel harmony to occur without compromising the autonomy of individual morphemes. In these languages, tones are superfluous because grammatical morphemes bear the primary functional load of linking morphemes into words, and words into longer constituents. Furthermore, word-final devoicing is not expected, as it would compromise the formal autonomy of morphemes. Aside from vowel harmony, morphophonemic processes, such as metathesis, umlaut, or any other process compromising the formal integrity or morphemes, are minimal. This lack of alternations maintains strict morphemic boundaries, which explains why complex consonant clusters are rarely formed/attested at the word edges in these languages. Turkish and Finnish are representative languages of this group and are discussed in turn in the following sections.

5.2.1 Turkish

Turkish is a Turkic language. Turkish is agglutinative. Words are built from stems + chains of suffixes**.** Each suffix usually carries a single grammatical function. An example from Turkish is given in (24).

(24)

An example (Turkish):

ev ‘house’

evler ‘houses’

evlerde ‘in the houses’

evlerinizde ‘in your houses’

Each suffix is easily segmentable:

-ler - plural

-de - locative (‘in’)

-iniz - your (possessive)

Vowel harmony restricts which vowels can co-occur in a word. Turkish has two types of progressive vowel harmonies. Suffixal vowels harmonize with root vowels based on front vs. back and rounded vs. unrounded features. Inkelas (2011:69) gives the concise description of backness harmony: “…a very general rule of progressive vowel harmony determines the value of [back] for the vowels of most suffixes, which surface with front vowels following roots whose final vowel is front (e.g., *gyl-ler* ‘rose-PL’, *anne-ler* ‘mother-PL’) but with back vowels following stems whose final vowel is back (e.g., *ok-lar* ‘arrow-PL’, *elma-lar* ‘apple-PL’). The process is not exceptionalness as for example, the associative suffix -*gil* retains its front vowel even when the base ends in a back vowel and the hypocoristic suffix -*o* retains its back vowel even when the truncated base has a front vowel (Lewis 2000, Göksel & Kerslake 2005; Kabak 2011). Bacanlı et al. (2020: 37) even suggest "… an analysis in which morphology and lexicon control VH and other non-automatic phonological processes, which are thus claimed to be no longer part of the active phonology of Turkish". See more examples of [±back] and [±round] vowel harmonies in (25a) and (25b) respectively.

(25)

a.	SG	PL
	ev 'house'	*evler* 'houses'
	okul 'school'	*okullar* 'schools'
b.	SG	POSS
	ked 'face'	*kedim* 'my face'
	köy 'village'	*köyüm* 'my village'

Typical syllable structure in Turkish is (C)V(C) (consonant (optional) + vowel + consonant (optional)). Turkish does not allow word-initial consonant clusters. However, loanwords with clusters are often adapted or kept intact in modern Turkish. Native Turkish words do not end in voiced stops (b, d, g). Allowed word-final consonants (in native words) are one of the following: {-k, -p, -t, -l, -n, -r, -m, -s, -ş}.

It is commonly acknowledged in linguistic research that Turkish features word-final devoicing (see Özçelik (2025). This issue is examined in Butskhrikidze (1994b), who, by analyzing both historical (beginning with Proto-Turkic) and contemporary data (from modern Turkic languages), argues that voiceless consonants at the ends of words in Modern Turkish and other Turkic languages are not the result of devoicing processes but are inherently voiceless. In contrast, voicing between vowels is a secondary development in these languages.

Turkish does not use accent contrastively. In most native Turkish words, primary accent falls on the last syllable.

Next, I turn to the Finnish language.

5.2.2 Finnish

Finnish belongs to the Uralic language family. Finnish is agglutinative. Words are built by adding suffixes to a root. Each suffix has one grammatical function (clear boundaries). It’s highly inflectional**,** especially in nouns and verbs**.** Finnish has a very weak word-initial accent and vowel harmony (back/front). No native noncompound word can contain vowels from the group {*a*, *o*, *u*} together with vowels from the group {*ä*, *ö*, *y*}.

According to Suomi et al. (2008), “Finnish vowels belong to one of three classes: the front harmonic /y/, /ø/, /æ/; the back harmonic /u/, /o/, /ɑ/; and the harmonically neutral /i/, /e/” (p.51)

Neutral vowels can appear in either front or back vowel words, but they don’t affect the harmony of suffixes.

> The major restriction is that, within an uncompounded word, vowels from the front harmonic and back harmonic classes cannot co-occur, while harmonically neutral vowels can co-occur with vowels from both harmonic classes. Thus there are words like kylä ‘village’ (only front harmonic vowels), talo ‘house’ (only back harmonic vowels), and isä ‘father’, kirja ‘book’, kesä ‘summer’, kello ‘clock’ (a mixture of harmonic and neutral vowels). Vowel harmony does not apply across the boundary between the components of compound words: iso+isä ‘grandfather’, kesä+loma ‘summer vacation’.
>
> Suomi et al. (2008:51)

In Finnish, the stem vowel determines the harmony class of the suffixes. “For example, the singular inessive of talo is talossa ‘in a/the house’, that of kylä is kylässä ‘in a/the village’” (Suomi et al. (2008:51).

Typical Finnish syllable structure is: (C)V(C). Finnish does not allow native words to begin or end in consonant clusters. Beisdes monomorphomic context, “In Finnish no MPT clusters result from affixation (ex+puoliso ‘former husband’) or compounding (syys+myrsky ‘autumn storm’), but MPT clusters are often simplified through consonant deletion or assimilation, as in the partitive last+ta of lapsi ‘child’, or the participle hakan+nut of hakata ‘to beat’ (Klaus Laalo, pers. comm.). Estonian, which is less agglutinating than Finnish (Skalička 1979), has more vowel loss and also more clusters (Skalička 1979: 308)” (Dressler et al. 2019, p.96).

The Finnish consonant inventory is limited. Some consonants (like b, d, f, g) only occur in loanwords**.** Consonant and vowel length are contrastive**.**

(26)

tuli ‘fire’

tulli ‘customs’

tuuli ‘wind’

Double consonants are allowed medially, but not word-initially, e.g., *soppa* ‘soup’ and *matto* ‘rug’. A glottal stop appears at certain morpheme boundaries, for instance, [ɑnnɑʔːollɑ] ‘let it be’, orthographically *anna olla*.

Traditionally, /b/ and /g/ were not counted as Finnish phonemes, since they appear only in loanwords. The status of /d/ is somewhat different from /b/ and /g/, since it also appears in native Finnish words, as a regular 'weak' correspondence of the voiceless /t/. Historically, this sound was a fricative, [ð], varyingly spelled as *d* or *dh* in Old Literary Finnish.

The phonemic template of a syllable in Finnish is (C)V(C)(C), in which C can be an obstruent or a liquid consonant. V can be realized as a doubled vowel or a diphthong. A final consonant of a Finnish word, though not a syllable, must be a coronal one; Standard Finnish does not allow final clusters of two consonants.

Accent is not contrastive and is always on the first syllable of the word in Finnish. Pitch and length are more important than accent in Finnish prosody. In longer words, secondary accent typically falls on every second syllable after the primary accent, starting from the third syllable, e.g., *tietokoneella* [ˈtie.to.ˌko.ne.el.la] ‘on the computer’.

Suomi, et.al. (2008) argues that “Suomi, McQueen & Cutler (1997) demonstrated that Finnish listeners can exploit harmonic mismatch information in an on-line speech segmentation task. For example, listeners found it easier to detect words like hymy at the end of the nonsense string puhymy (where there is a harmony mismatch between the first two syllables) than in the string pyhymy (where there is no mismatch). Similarly, palo was detected easier in kypalo than in kupalo” p.54.

5.2.3 Summary

By examining word boundary markers in Turkish and Finnish, we can observe that vowel harmony and weak accent function as indicators of word boundaries in these languages. Therefore, the following for agglutinative languages[21] can be concluded:

[21] I recognize that there are many agglutinative languages, for instance, Bantu languages that have agglutinative morphology and no vowel harmony, e.g., Zulu and Shona. There is an important difference between Altaic agglutinative and Bantu agglutinative languages in word structure. In Altaic languages lexical morpheme (stem) coincides with a word boundary and thus it is formally independent, while in Bantu languages word = prefix + stem. Because the stem is formally dependent in Bantu languages, it cannot govern a process such as vowel harmony, and instead we find tones in these languages having grammatical function. It will need a separate paper to explore boundary markers in Bantu languages that have a very diverse morphological and prosodic patterns within the language family.

(27) Boundary markers in agglutinative languages[22]

- ✓ Vowel harmony
- ❑ Very weak word accent
- ❑ Not many consonant co-occurrence restrictions within stems
- ❑ No obstruent clusters (exceptions in loanwords)
- ❑ no final devoicing

- ❖ Medial voicing is a phenomenon worth studying in Finnish and Turkish

5.3 Fusional languages

Fusional languages (also called inflectional languages) are a type of synthetic language characterized by the way they encode multiple grammatical features into single affixes or word endings. Here are the key characteristics of fusional languages: a) single morphemes encode multiple grammatical categories (e.g., in Spanish *habló* ‘he/she spoke’, the suffix *-ó* (3rd person, singular, past tense, indicative mood, all are fused into one suffix), b) the boundaries between morphemes are not always clear, d) rich inflectional morphology. Blurring of morpheme boundaries is compensated by increase in word boundary markers: both suprasegmental and segmental. In fusional languages, from word boundary markers I expect word-final devoicing, consonant clusters at the word edges and fixed accent. In contrast to vowel harmony in

[22] Some North Caucasian languages are highly agglutinative and have vowel harmony (for more on this, see Butskhrikidze, 2024).

agglutinative languages, fusional languages feature the morphophonological process of umlaut. Two representative languages of this group are Albanian and Georgian, discussed in turn.

5.3.1 Albanian

Albanian, a member of the Indo-European language family, is classified as a fusional language. To illustrate its fusional characteristics, I will examine the highly versatile morpheme *të*.

In Albanian, *të* can be analyzed in two very different ways depending on the construction: a) as a homophonous form or b) as a portmanteau morpheme **(**Newmark, Hubbard, & Prifti (1982)**.**

të refers to multiple distinct morphemes that happen to share identical pronunciation and spelling: a) subjunctive particle (e.g., as in *dua të shkoj* ‘I want to go’; b) as part of optative verb, expressing a strong wish, blessing, or a curse, e.g,. *të lumtë goja*! ‘Well said!’ (literally: may your mouth be blessed!); c) linker/article (also called adjectival article), e.g,. *librat të mire* ‘good books’. Here *të* is used when the noun it refers to is masculine plural, feminine plural and only in the dative or genitive cases. With genitive nouns *të* expresses possession, e.g., in *shtëpia e prindërve të mi* ‘the house of my parents’. In this example, *të* besides showing possession, it simultaneously expresses definiteness. Thus, in last two cases it is a portmanteau morpheme (conveying multiple meanings). *të* can also be a personal pronoun. Specifically, it is the pronominal clitic for the second person singular (‘you’, accusative or dative, e.g., *unë të shoh ty* ‘I see you’, literally ‘I you (clitic) see you’, *profesori të dha një libër* ‘The professor gave you a book’ literally ‘the professor to you (clitic**)** gave a book)) or the third person plural (‘them’, dative, e.g., *thuaju të vijnë* ‘tell them to come’). When the pronoun *të* (to you) meets the 3rd person accusative pronouns *e* (him/her/it) or *i* (them), they fuse together:

(28)

Dative Clitic + Accusative Clitic = Fused Clitic Combination
të *(to you)* + e *(him/her/it)* = ta

ta dhashë ‘I gave it to you’

të *(to you)* + i *(them)*= ti

ti dhashë librat ‘I gave the books to you’

Verbs, nouns, adjectives, and pronouns are inflected. A single suffix can encode tense, mood, number, or person. It has both prefixal and suffixal morphology.

Unlike many languages, definiteness is marked on the noun itself using suffixes:

(29)

libër ‘a book’ *libri* ‘the book’

vajzë ‘a girl’ *vajza* ‘the girl’

zog ‘a bird’ *zogu* ‘the bird’

The root consonant might change when a suffix is added to it. For instance, *zog* [zɔg] ‘bird’ and *zogjtë* [zɔɟtə] ‘the birds’.

Concerning phonotactic rules, in some cases, final obstruent devoicing is attested.

Consonant clusters, maximally CCC, occur both word-initially and word-finally.

(30)

shpresa [ˈʃprɛsa] ‘hope’ *ankth* [ˈaŋkθ] ‘anxiety’

shkretëtirë [ʃkrɛtəˈtirə] ‘desert’

mbreti [ˈmbrɛti] ‘king’

The Consonants in clusters are strictly constrained and predictable, for instance in the word initial #C1C2C3, C1 can only be a nasal or fricative (e.g., *m*, *n*, *s* or *ʃ*), C2 can only be a stop (e.g., *b*, *p*, *t* or *g*), while C3 can only be an approximant (e.g., *r*, *l* or *j*). In this respect, Albanian word initial clusters resemble the English ones. In Albanian, the repertoire is much richer and varied, while English has only a very limited subset.

Dynamic 'free' word accent is mainly on the penultimate syllable (there are a very few instances when accent is distinctive, e.g., *bari* [baˈɾi] 'shepherd' vs. *bari* [ˈbaɾi] 'grass'). Accent often shifts in derived or inflected forms (for more on Albanian accent placement see Trommer (2013)).

5.3.2 Georgian

Georgian belongs to the Kartvelian (South Caucasian) language group. It has fusional morphology (both prefixal and suffixal). A single affix can express multiple grammatical meanings. For instance,

(31)

მოგწერე [mogts'ere] 'I wrote to you'

მო- (*mo-*) 'version prefix'

გ- (*g-*) '2nd person object'

წერ (*ts'er*) 'root "write"'

ე (-*e*) ‘1st person subject, aorist tense’

Consider /v/ metathesis in Georgian. Butskhrikidze (2002) claims that “/v/, which is part of the thematic suffix in verb forms, violates integrity of the root when followed by the infinitival suffix /-a/. For example, the third person singular of the root /xar/ ‘to gnaw’ is /xr-av-s/ (with deletion of the root vowel). The infinitival form, however, is /xvr-a/ (with deletion of both the root vowel and the vowel in the thematic suffix), where the thematic suffix consonant /v/occurs between the two root consonants” p.93. Illustrative examples are given in (32).

(32)

Regular metathesis in verb forms

ROOT	PRES 3rd SG	INF	
	(/-av/ THEM SUFF)	(/-a/ INF SUFF)	
xar	xr-av-s	xvr-a	‘to gnaw’
k’ar	k’r-av-s	k’vr-a	‘to tie’
xan	xn-av-s	xvn-a	‘to plough’
k’al	k’l-av-s	k’vl-a	‘to kill’
sxal	sxl-av-s	sxvl-a	‘to chop off’
jer	jr-av-s	jvr-a	‘to move’

Butskhrikidze 2002: 94

Concerning word-final devoicing in Georgian Butskhrikidze (2002) claims that “Since no minimal word ends in a consonant in Georgian, devoicing occurs in grammatical affixes, e.g., /v-a-k’et-eb/ is pronounced as [vak’etep] ‘I do something’, where /b/ belongs to the thematic suffix /-eb/ and /k’aʦ -ad/ is pronounced as [kaʦ -at] ‘man ABL’, where /-ad/ is the Ablative case marker. Devoicing does not occur when the suffixes are followed by another suffix, for instance in forms such as /v-a-k’et-eb-di/ [v-a-k’et-eb-d-i] ‘I was doing’, and /k’aʦ -ad-a/ [k’aʦ -ad-a] ‘like a man’” (p 85).

Butskhrikidze (2002) study of word-initial consonant sequences has shown that the complexity of consonant sequences is quite deceptive in Georgian.

> The two-consonant clusters which seem to be the building blocks of much longer consonant sequences are of the type obstruent + sonorant. Thus, they largely obey the SSP. The most frequently realized clusters are combinations of obstruent + sonorant, harmonic clusters and consonants that share a laryngeal specification and have [front] + [back] place of articulation. Clusters that do not obey the SSP are largely of morphological origin.
>
> Butskhrikidze (2002:112)

The same transparency is observed in long consonant sequences (e.g., attested in words such as *prckvna*, *brdɣvna*, etc.). The following generalization is suggested by Gvinadze (1970):

(33)

I	/b p p’ m/
II	/r/
III	/d t t’ ʣ ʦ ʦ’ ʤ ʧ ʧ’ z s ʒ ʃ /
IV	/g k k’ ɣ x χ’/
V	/v/
VI	/r l m n/

(cited in Butskhrikidze 2002:108)

Word accent in Georgian is not contrastive. It is phonetically subtle (not accompanied by strong pitch, loudness, or length changes). Georgian almost always stresses the penultimate syllable (see Akhvlediani 1949).

5.3.3 Summary

By examining word boundary markers in Albanian and Georgian, One can observe that word boundaries are more varied and richer compared to isolating and agglutinative languages. Word-final devoicing, word accent and consonant clusters are only some of the word boundaries attested in these languages. Umlaut/ablaut are not attested in Albanian or Georgian, but one could potentially except this phenomenon to exist in a fusional language. Therefore, we can conclude the following for fusional languages:

(34) Boundary markers in fusional language

- ✓ Fixed/free word accent
- ✓ many consonant co-occurrence restrictions, e.g., consonant clusters at the word edges
- ✓ Final devoicing or other signals at the word-edges
- ❖ Interestingly, in these languages, instead of vowel harmony, we find umlaut/ablaut.

6 Conclusions

Word boundary markers are a perfect example of how syntagmatics and paradigmatics are intertwined in a language. They play a crucial role in ensuring the autonomous nature and internal coherence of words. The type and frequency of these markers are influenced by a language's morphological structure, with isolating, agglutinative, and fusional languages exhibiting distinct patterns.

The findings reveal a strong relationship between a language's morphological type and its use of word boundary markers. Isolating languages, which lack inflectional morphology, rely heavily on tones and syllable structure to signal word boundaries. Agglutinative languages, which use extensive affixation, employ vowel harmony and weak word accent. Fusional languages, which combine multiple morphemes into single words, often use fixed or free accent systems, consonant clusters, and final obstruent devoicing. The findings are summarized in (35).

(35) Morphological typology and word boundary markers

Type of a morpheme / Type of a language	Lexical morphemes	Grammatical morphemes	Word boundary markers
Isolating	Independent	Independent	Tones, ‘initials’ and ‘finals’ (mutually exclusive sets of sounds attested at the word-edges)
Agglutinative	Independent	Dependent	Vowel harmony (usually stem-controlled)
Fusional	Dependent	Dependent	Fixed accent, Free accent, Final obstruent devoicing, Consonant clusters (obstruent clusters)

Word boundary markers in all these languages guarantee the autonomous nature of words and facilitate speech comprehension. The aim is one, but the strategies how it is achieved varies from language to language, depending on its morphological type/makeup.

As also demonstrated in earlier studies in Butskhrikidze (2022), the formal dependence of both lexical and grammatical morphemes entails the existence of the largest (the most varied) number of word boundary markers in fusional languages. The great uniformity of morpheme-types in isolating languages entails a limited number of word boundary markers. Agglutinative languages stand somewhere in between in this respect.

Given our sample size of six languages, the quantitative trends identified herein remain exploratory and cannot be generalized as universal linguistic typologies. To statistically validate these preliminary patterns, future research must scale up this methodology by testing it against a significantly larger, typologically diverse dataset. Additionally, subsequent research should also explore the interplay between phonology, morphology, and language processing, particularly in the context of language contact and change. The questions that could be asked are the following:

what happens when fusional and isolating languages are in sustained contact? Do boundary strategies shift?

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